

Example Problem

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Exercise

Sakurai 5.29

Problem Statement

Consider a composite system made up of two spin $\frac{1}{2}$ objects. For $t < 0$, the Hamiltonian does not depend on spin and can be taken to be zero by suitably adjusting the energy scale. For $t > 0$, the Hamiltonian is given by

$$H = \left(\frac{4\Delta}{\hbar^2} \right) \mathbf{S}_1 \cdot \mathbf{S}_2.$$

Suppose the system is in $|+-\rangle$ for $t \leq 0$. Find, as a function of time, the probability for its being found in each of the following states $|++\rangle$, $|+-\rangle$, $|-\rangle$, and $|--\rangle$:

- By solving the problem exactly.
- By solving the problem assuming the validity of first-order time-dependent perturbation theory with H as a perturbation switched on at $t = 0$. Under what condition does (b)

An Exact Solution

To solve this exactly, we need to work with states that are diagonal in the perturbing Hamiltonian, so that we may time evolve it. We have worked this system to the bone, and we know that the eigenbasis constitutes of linear combinations of the $|\pm\pm\rangle$ states. To use this fact, we may produce the rotated states, following the book's convention.

$$\begin{aligned} |1, 1\rangle &= |++\rangle, & |1, 0\rangle &= \frac{1}{\sqrt{2}} (|+-\rangle + |-+\rangle), \\ |1, -1\rangle &= |--\rangle, & |0, 0\rangle &= \frac{1}{\sqrt{2}} (|+-\rangle - |-+\rangle) \end{aligned}$$

We identify the initial condition as $|+-\rangle = \frac{1}{\sqrt{2}}(|1, 0\rangle + |0, 0\rangle)$. The perturbing spin Hamiltonian is

$$H = \frac{4\Delta}{\hbar^2} \mathbf{S}_1 \cdot \mathbf{S}_2 = \frac{4\Delta}{\hbar^2} \left(S_1^z S_2^z + \frac{1}{2} S_1^+ S_2^- + \frac{1}{2} S_1^- S_2^+ \right) \quad (1)$$

We identify the initial conditions as follows: $c_{10}(0) = c_{00}(0) = \frac{1}{\sqrt{2}}$. We now follow an explicit time evolution:

$$e^{-iHt/\hbar} |+-\rangle = \frac{1}{\sqrt{2}} e^{-iHt/\hbar} (|1, 0\rangle + |0, 0\rangle) = \frac{1}{\sqrt{2}} \left(e^{-i\Delta t/\hbar} |1, 0\rangle + e^{3i\Delta t/\hbar} |0, 0\rangle \right) \quad (2)$$

This comes from the spin calculations done with the expansion in eq.(1):

$$\frac{1}{\hbar^2} S_1 \cdot S_2 |+-\rangle = -\frac{1}{4} |+-\rangle + \frac{1}{2} |-+\rangle \quad (3)$$

$$\frac{1}{\hbar^2} S_1 \cdot S_2 |-+\rangle = -\frac{1}{4} |-+\rangle + \frac{1}{2} |+-\rangle \quad (4)$$

which allowed us to determine the results for $S_1 \cdot S_2 \frac{1}{\sqrt{2}} (|+-\rangle \pm |-+\rangle)$. Now, we note that we may express the results $c_{10}(t) = \frac{\exp(-i\Delta t/\hbar)}{\sqrt{2}}$ and $c_{00}(t) = \frac{\exp(3i\Delta t/\hbar)}{\sqrt{2}}$. This implies that $c_{11}(t) = c_{++}(t) = 0$ and $c_{1-1}(t) = c_{--}(t) = 0$, since $|1, 1\rangle = |++\rangle$ and $|1, -1\rangle = |--\rangle$. Now we decompose the $|1, 0\rangle$ and $|0, 0\rangle$ back into the $|\pm\mp\rangle$ states, to get

$$c_{+-}(t) = \frac{1}{2} \left(e^{-i\Delta t/\hbar} + e^{3i\Delta t/\hbar} \right), \quad c_{-+}(t) = \frac{1}{2} \left(e^{-i\Delta t/\hbar} - e^{3i\Delta t/\hbar} \right).$$

We may now square the coefficients to get the probabilities:

$$P(+-, t) = \cos^2 \left(\frac{2\Delta t}{\hbar} \right), \quad P(-+, t) = \sin^2 \left(\frac{2\Delta t}{\hbar} \right), \quad P(++ , t) = P(-- , t) = 0 \quad (5)$$

The Next Best Thing

Our primary goal is to employ

$$c_m(t) = \delta_{nm} - \frac{i}{\hbar} \int_0^t dt' e^{i\omega_{mn}t'} V_{mn}(t'), \quad (6)$$

to obtain the first order transition amplitudes.

While the Hamiltonian is not diagonal in the $\pm$ state basis, we note that the $|\pm\mp\rangle$ states form a closed subspace under the operation of the Hamiltonian. Thus, $\langle\pm\pm|H|\pm\mp\rangle$ vanishes identically, implying that, to first order, there are no allowed transitions to these states. We then only concern ourselves, to first order, with the transition $|+-\rangle \rightarrow |-+\rangle$. We use eq.(6) to compute this. The matrix element $\langle -+|H|+-\rangle$ can be computed by noting that eq.(3) and the orthonormality of the 2-spin states implies $\langle -+|S_1 \cdot S_2|+-\rangle = \hbar^2/2$. Thus, the matrix element is simply 2Δ . However, to compute the exponential term, we need to work in the diagonal eigenbasis of the Hamiltonian once more. This consideration can be simplified by noting that $|+-\rangle$ and $|-+\rangle$ are simply the positively ordered symmetric and anti-symmetric combinations of $|1, 0\rangle$ and $|0, 0\rangle$. Thus, the energy difference between these states is simply 0, given that they are eigenstates of the Hamiltonian. This can be explicitly verified by noting that eq.(3) and eq.(4) imply that $\langle +-|S_1 \cdot S_2|+-\rangle = \langle -+|S_1 \cdot S_2|-+\rangle$. Then we may state the transition amplitude:

$$c_{-+}(t) = -\frac{i}{\hbar} \int_0^t dt' 2\Delta \quad (7)$$

This is a trivial integral and provides the result $c_{-+}(t) = \frac{-2i\Delta}{\hbar}t$. Since we don't simply desire the amplitudes for transition, but wish to verify the validity of "staying in place" with perturbation theory, we compute (with the obvious $\omega_{(+-)(+-)} = 0$):

$$c_{+-}(t) = 1 - \frac{i}{\hbar} \int_0^t dt' (-\Delta) \quad (8)$$

This allows us to present the probabilities:

$$P^{(1)}(+-, t) = 1 - \left(\frac{\Delta t}{\hbar} \right)^2, \quad P^{(1)}(-+, t) = \left(\frac{2\Delta t}{\hbar} \right)^2, \quad P^{(1)}(++ , t) = P^{(1)}(-- , t) = 0 \quad (9)$$

Note that $P^{(1)}(-+, t) \approx P(-+, t)$ for $t \ll \frac{\hbar}{2\Delta}$, but $P^{(1)}(+-, t)$ does not maintain its validity even for short times, implying the non-conservation of probability. So this would be a useful tool to compute the transition probability for short times.